

Customer Churn — Exploratory Data Analysis

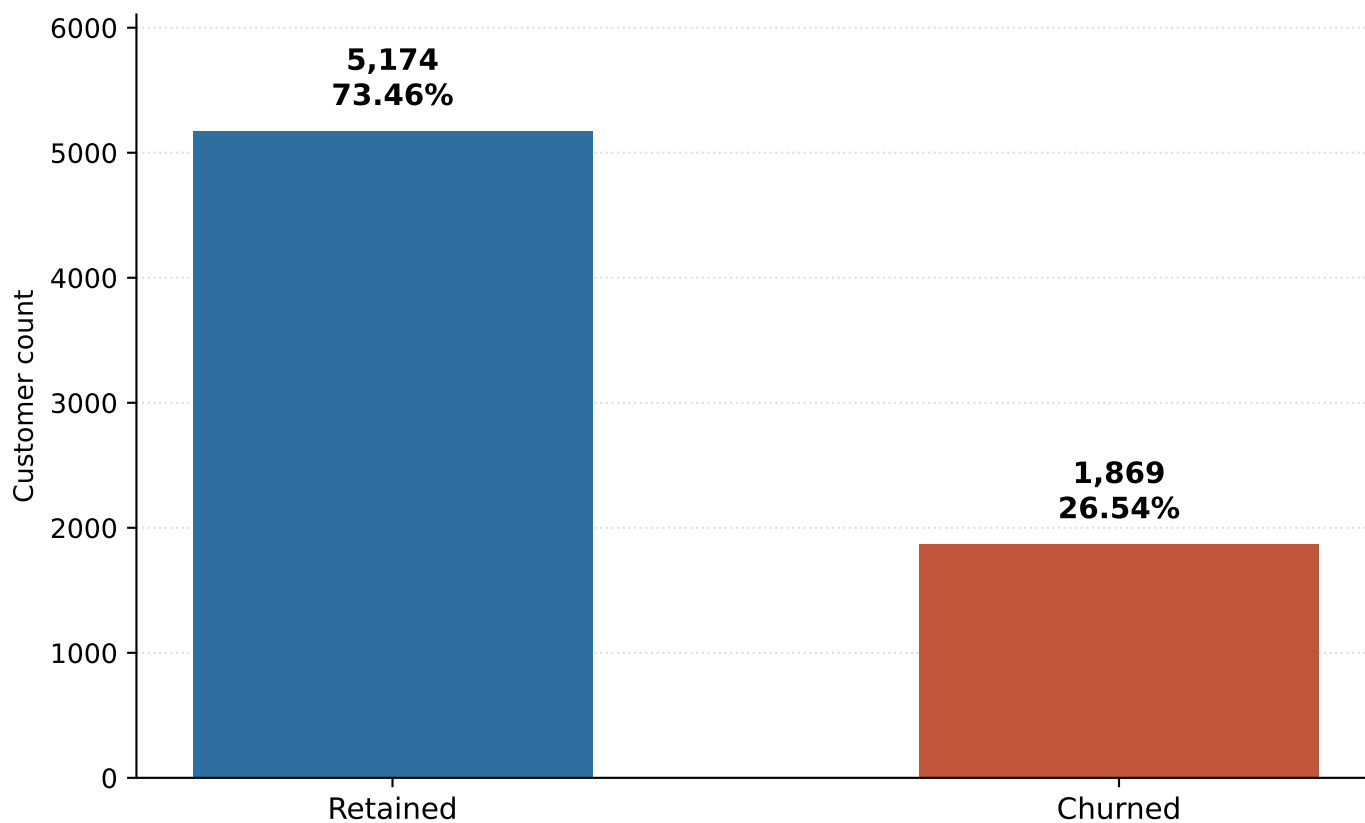
1. Dataset Overview

Dataset:	data/raw/WA_Fn-UseC_-Telco-Customer-Churn.csv
Total records:	7,043
Target variable:	Churn
Analysis scope:	Investigation 1 – retention vs. churn baseline Distribution of the target variable only. No other variable is examined in this report.
Data treatment:	Read-only. The raw dataset is not modified, cleaned, or transformed by this analysis.

2. Retention vs. Churn

Counts and percentages are computed from the loaded dataset.

Customers by status



Customer Status	Customer Count	Percentage
Retained	5,174	73.46%
Churned	1,869	26.54%
Total	7,043	100.00%

3. Interpretation

Of the 7,043 customers in the dataset, 5,174 customers (73.46%) are recorded as retained (Churn = "No") and 1,869 customers (26.54%) are recorded as churned (Churn = "Yes").

Retained customers therefore outnumber churned customers by 3,305 customers, a gap of 46.93% of the dataset.

Every record carries one of the two documented target values, "No" or "Yes"; the target contains no missing values and no other category. The two percentages above therefore account for the whole dataset.

These proportions are the baseline for the exploratory analysis that follows. Subsequent investigations compare the churn rate within individual segments of the data against the overall rate reported here, and the same proportions describe the class balance any later model would be trained against.

This investigation examines the distribution of the target variable only. It establishes how many customers churned, not why: no relationship between churn and any other variable has been measured at this stage, and no cause is inferred from these counts.